

Assignment#1

Deadline: 18th sept 2026

Question 1

Choose one app that you use regularly, such as WhatsApp, YouTube, Google Maps, Instagram, Spotify, or Microsoft Word.

Identify **three different interaction styles** used in the app. You may consider styles such as direct manipulation, menu-driven interaction, gestures, command-based input, voice control, or agent-based interaction. The three interaction styles should be different. For each style, give one real example from the selected app.

For example, in Google Maps, tapping and moving the map is an example of direct manipulation, while selecting options from the menu is menu-driven interaction.

Next, choose **one specific task** in the same app, such as sharing a photo, finding a location, playing a song, or creating a document.

Evaluate this task using the following two points:

- **Execution:** Is it clear what the user needs to do to complete the task? Explain what makes the task easy, difficult, or confusing.
- **Evaluation:** After performing the task, can the user easily understand whether the task was completed successfully or not? Explain what feedback the app provides.

Finally, search **Google Scholar** for one research paper related to the usability of your selected app or a similar application. Read the abstract and briefly explain the main finding of the research. Then explain in your own words how this finding relates to good HCI and interactive system design.

Submission: Submit a **1 - 2 page report**. Include the **title and reference/citation of the research paper** you used.

Question 2

Choose a physical machine or kiosk that people commonly use, such as an ATM, self-checkout machine, ticket vending machine, hospital check-in kiosk, parking payment machine, or smart home control panel.

First, describe the machine, who normally uses it, and what users usually want to achieve.

Then explain the user's experience from start to finish:

1. What does the user want to achieve?
2. What does the user plan to do?
3. What action does the user perform?
4. How does the machine respond?
5. What information does the machine display?

6. Can the user understand the information?
7. How does the user know whether the task was successful or unsuccessful?

Identify any confusing, difficult, or frustrating points in the interaction. For example, the user may have difficulty finding a button, understanding an instruction, or knowing what to do next.

Next, identify **at least three physical or ergonomic problems**. These may include small buttons, difficult-to-read text, poor colour choices, inconvenient screen height, poor lighting, screen reflections, or difficult positioning.

Finally, suggest **two improvements** to the machine. For each improvement, explain:

- What exactly should be changed?
- How would the change improve the user's experience?
- Which human ability or limitation does it address, such as vision, memory, attention, perception, or movement?

Question 3

Choose a real-world problem that people face in an area such as healthcare, education, smart cities, accessibility, transportation, or the environment.

Design a **new interactive system** that could solve this problem using modern or future technology.

First, clearly describe:

- What exact problem does the system solve?
- Who will use the system?
- How will users provide input? For example, touch, voice, gestures, sensors, or cameras.
- How will the system provide output? For example, screen, sound, vibration, or lights.
- What interaction style will mainly be used?

Connect your design to **one or two classic HCI interaction styles**, such as direct manipulation, ubiquitous computing, agent-based interfaces, or command-based interaction.

Explain how your system uses these interaction styles and **what makes your idea different or better than existing solutions**.

Then discuss the following:

- **Privacy/Ethics:** Identify **two possible privacy or ethical risks**, such as unauthorized tracking, data leaks, or biased decisions.
- **Technology Limitation:** Identify **one technology limitation** that could make the system difficult to develop, such as sensor accuracy, battery life, processing power, or internet connectivity.

- **Accessibility:** Explain how people with disabilities will be able to use your system. Consider features such as large text, voice control, captions, vibration alerts, simple navigation, or high-contrast displays.

Finally, create a simple **4–5 panel storyboard** showing:

1. The user facing the problem.
2. The user starting to use the system.
3. The system responding to the user.
4. The user receiving the result.
5. An optional final panel showing the successful outcome.

You may draw the storyboard by hand and take a clear photograph, or create it using a digital drawing tool.